

and symmetric peripheral gangrene occur more frequently in infants and children than in older individuals.² In neonates, purpura fulminans can be a manifestation of inherited homozygous protein C deficiency or rarely of protein S deficiency. BSIs in older children with these underlying deficiencies (and properdin deficiency, splenic dysfunction, or neutropenia) can lead to purpura fulminans.

Palpable Purpura

Unlike petechiae and ecchymoses, palpable purpura is a condition in which raised, purpuric papules and plaques can range from a few millimeters to a few centimeters in diameter. Palpable purpura favors dependent areas such as the lower extremities, but in the supine patient, lesions can occur on the back, buttocks, and distal arms. Palpable purpura can evolve into nodular, vesicular, pustular, necrotic, or ulcerative lesions.

Palpable purpura classically is associated with leukocytoclastic vasculitis, an inflammatory process that injures the vessel.³³ Histopathologically, fibrin deposition in vessel walls, neutrophilic debris, and perivascular lymphocytes are seen. Palpable purpura also can occur with *S. aureus* BSI or in bacterial endocarditis as the result of infectious microemboli occluding the vascular lumen. Osler nodes and Janeway lesions found in bacterial endocarditis are types of microembolism-induced palpable purpura. Osler nodes are tender nodules that are seen on the palms, soles, and pads of the fingers and toes. Janeway lesions are nontender, purpuric macules, papules, or nodules usually found on the palms and soles. Lesions similar to Osler nodes have been reported in typhoid fever, polyarteritis nodosa, systemic lupus erythematosus, Churg-Strauss syndrome, mixed cryoglobulinemia, and Wegener granulomatosis.^{1,2}

Palpable purpura has a variety of causes, including infection (10%–15%), drug reactions, immunization, and autoimmune disease. The most common infectious causes include *N. meningitidis*, *S. aureus*, and *N. gonorrhoeae*, but it also has been caused by *Mycobacterium*, *Rickettsia*, and *Mycoplasma* spp., and rarely by *Bartonella*, *Salmonella*, *Campylobacter*, *Yersinia*, *Brucella* spp., and *Treponema pallidum*. Viral causes include hepatitis viruses (i.e., hepatitis C is more common than hepatitis B, and hepatitis B is more common than hepatitis A), hepatitis virus vaccine, and infection due to human immunodeficiency virus (HIV), parvovirus, SARS-CoV-2, and rarely by cytomegalovirus, varicella, and influenza.¹

Noninfectious causes of palpable purpura in the pediatric population include Henoch-Schönlein purpura (HSP), the most common vasculitis seen in childhood, and acute hemorrhagic edema of infancy (AHEI). Palpable purpura is the presenting sign in almost 50% of cases with HSP. Diagnostic criteria for HSP include palpable purpura (with lower limb

predominance), a mandatory criterion, along with at least one of the following: diffuse abdominal pain, tissue biopsy showing predominant immunoglobulin A (IgA) deposition, acute arthritis or arthralgia of any joint, and renal involvement.³⁴ If the distribution of cutaneous lesions is atypical, biopsy confirming IgA deposition may be required. Hemorrhagic bullae also have been reported in HSP, and their presence does not appear to affect prognosis.³⁵

HSP is most common between the ages of 5 and 10 years, commonly following an upper respiratory tract infection, and AHEI usually manifests in children younger than 2 years of age.³⁶ Some consider AHEI a distinct entity, whereas others consider it infantile HSP. AHEI typically manifests with fever, edema, and rosette-shaped, purpuric cutaneous lesions favoring the face, ears, and limbs. AHEI has a benign course, resolving over 3 weeks without sequelae, and it lacks visceral involvement.³⁷

Exercise-induced purpura is a recently reported benign vasculopathy, associated with warm weather and prolonged activity, consisting of purpura extending above the sock line to the knees.³⁸ Rumpel-Leede phenomenon, initially reported in association with scarlet fever, is another benign acute purpuric manifestation, which can occur in healthy neonates after prolonged exposure to excessively tight baby carriers, due to its tourniquet-like forces.³⁹ These manifestations can be mistaken for HSP or other pathogenic causes of purpura; thus, a careful history is needed to avoid unnecessary workup and evaluation.

Retiform Purpura

Retiform purpura results from occlusion of dermal and subcutaneous vessels, which causes a lace-like pattern of purpura. A complete reticulate pattern is not usually seen, but rather a puzzle piece–like or branching pattern occurs because of the angulated or sometimes serpentine pattern of the purpura. Vascular invasion by fungi (e.g., *Mucor*, *Aspergillus* in immunocompromised patients), gram-negative organisms (e.g., *Pseudomonas*, the most common cause of ecthyma gangrenosum) that multiply in arterioles, and disseminated *Strongyloides* can cause noninflammatory retiform purpura predominantly in the periumbilical region with an appearance likened to thumbprints, whereas vasculitis during BSI can cause inflammatory retiform purpura.¹ Of the recently reported cutaneous manifestations of Covid-19, a type of retiform purpura, known as the *livedo reticularis subtype*, has been documented; further studies are needed to better understand the pathogenesis of this disease.²³

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71 Urticaria and Erythema Multiforme

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Urticaria and erythema multiforme (EM) are common cutaneous hypersensitivity reactions seen in children. They can be triggered by infection, although noninfectious causes also are prevalent. Although distinctly different entities, acute urticaria often is misdiagnosed as EM. Although urticaria usually is an isolated finding, urticaria and urticarial dermatoses can be seen in several syndromes and disorders such as papular urticaria, urticarial vasculitis, and the cryopyrin-associated periodic syndromes (CAPSs).

URTICARIA

Etiologic Agents

Acute and chronic forms of urticaria have been attributed to numerous bacterial, viral, fungal, and other infectious agents (Box 71.1).

Noninfectious causes and associations include foods, medications, physical stimuli, aeroallergens, and systemic diseases (Box 71.2).¹

In acute urticaria in children, an infectious cause can be demonstrated in up to 60% of cases.^{2–4} Viral upper respiratory tract and gastrointestinal (GI) tract infections are the primary infectious triggers of acute urticaria in children, with rhinovirus, adenovirus, rotavirus, and *Streptococcus pyogenes* the most commonly associated infectious agents.^{4,5}

Most cases of chronic urticaria in children have idiopathic, autoimmune, or physical causes.^{6–11} Physical urticaria is the most commonly identified cause. Triggers include cold exposure, pressure, heat, exercise, vibration, and water.⁹ Chronic urticaria can be associated with a variety of infections.^{2,7,10,12,13} The relevance of occult dental, sinus, or ear infections remains controversial.¹⁴ In one study of adults and children with chronic urticaria evaluated at an academic medical center in Turkey, 10 (31.2%)

BOX 71.1 Infectious Agents Associated With Urticaria**BACTERIA**

Borrelia burgdorferi (erythema chronicum migrans)
Chlamydomphila pneumoniae
Escherichia coli
Helicobacter pylori
Neisseria gonorrhoeae
Neisseria meningitidis
Pseudomonas aeruginosa
Shigella sonnei
Streptococcus pyogenes^a
Yersinia enterocolitica

FUNGI

Candida albicans
Cladosporium spp.
Coccidioides immitis
Histoplasma capsulatum
Candida glabrata
Trichophyton spp.

HELMINTHS

Ancylostoma duodenale
Anisakis simplex
Ascaris lumbricoides
Echinococcus spp.
Enterobius vermicularis
Fasciola hepatica
Necator americanus
Onchocerca volvulus
Schistosoma spp.
Strongyloides stercoralis
Toxocara canis
Trichinella spiralis

Trichobilharzia spp. (avian blood flukes)
Wuchereria bancrofti

MYCOPLASMA

Mycoplasma pneumoniae

PROTOZOA

Blastocystis hominis
Entamoeba histolytica
Giardia lamblia
Plasmodium spp.
Trichomonas vaginalis

RICKETTSIA

Coxiella burnetii

TREPONEMES

Treponema pallidum

VIRUSES

Adenovirus^a
 Coxsackieviruses A9, A16, B4, B5^a
 Cytomegalovirus
 Echovirus 11^a
 Epstein-Barr virus^a
 Hepatitis viruses A, B, C
 Influenza B virus^a
 Human immunodeficiency virus
 Measles virus, attenuated
 Mumps virus
 Parvovirus B19
 Respiratory syncytial virus^a

^aMost common.

of 32 of children with chronic urticaria tested positive for *Helicobacter pylori* antigen in their stool, and urticaria resolved after eradication of *H. pylori*.¹⁵ Based on the lack of resolution of most cases of chronic urticaria after treatment of the purported infectious trigger, however, some experts argue that there is no relationship between chronic infection and chronic urticaria.¹⁶ Some evidence suggests that patients with chronic autoimmune urticaria can manifest an infection-associated autoreactive response.¹⁷ There is no demonstrated association between chronic urticaria and malignancy.¹⁸

Delayed urticaria to red meat ingestion due to an immunoglobulin E (IgE) reaction to the oligosaccharide galactose- α -1,3-galactose (α -gal) found in some tick species has been identified and reported recently in children.¹⁹ Bites from the *Amblyomma americanum* tick in the US (and *Ixodes* species elsewhere) cause sensitization to α -gal, resulting in delayed urticaria, angioedema, and anaphylaxis after ingestion of mammalian meat.

Epidemiology

Urticaria is a common problem over a lifetime, occurring in 20%–25% of the population.²⁰ For children and adolescents, prevalence estimates range from 2%–20%.^{2,21–23} There is no known sex, ethnic, or racial predisposition. Most pediatric cases of urticaria are acute and self-limited. In children, chronic urticaria is uncommon, with a prevalence of only 0.1%–0.3%.²¹ Although controversy exists as to an increased prevalence of urticaria among people with a personal or family history of atopic disease, a prospective birth cohort study from Germany demonstrated a significant association between the cumulative prevalence of urticaria at age 10 years and allergic sensitization with peanut, soy, and wheat but not with aeroallergens. A personal history of asthma, eczema, and seasonal allergies and

parental history of atopy or urticaria also were strongly associated.²² A history of atopy has been associated with increased severity and duration of acute urticaria in children.^{12,24}

Pathogenesis and Pathologic Findings

Urticaria results from degranulation of mast cells and basophils with release of histamine and other vasoactive mediators.²⁵ Histamine acts through H₁ and H₂ receptors to produce vasodilation and altered vascular permeability, which results in fluid extravasation and dermal edema. Together with the axonal reflex, which produces the associated cutaneous flare, these features comprise the triple response of Lewis. Intense pruritus also can result from histamine released into the dermis. Other mediators of urticaria include kinins, endogenous opioids, prostaglandins, leukotrienes, cytokines, chemokines, and neuropeptides.

Urticaria can have an immunologic or nonimmunologic pathogenesis. Immune-mediated urticaria results from IgE-mediated type I hypersensitivity reaction; from the development of IgG autoantibodies against IgE or the high-affinity IgE Fc ϵ RI or Fc ϵ RII receptor (i.e., type II hypersensitivity reaction); from the binding of circulating immune complexes to mast cells expressing Fc receptors for IgG and IgM (i.e., type III hypersensitivity reaction); and rarely from T-lymphocyte-mediated activation of mast cells (i.e., type IV hypersensitivity reaction).²⁵

In nonimmunologic (pseudoallergic) urticaria, direct degranulation of mast cells and basophils occurs in response to exogenous factors such as pressure and cold; from activation of membrane receptors involved in innate immunity (e.g., by opioids); through direct mast cell toxicity, as occurs with iodinated radiographic contrast agents, drugs (e.g., aspirin,

ARTHROPOD BITES (PAPULAR URTICARIA)

Ants (*Solenopsis saevissima*)
 Bedbugs (*Cimex lectularius*)
 Bees^a
 Body lice (*Pediculus humanus*)^a
 Caterpillars
 Fleas (*Pulex irritans*)^a
 Chiggers (*Trombicula irritans*)
 Flies
 Gypsy moths
 Kissing bugs (*Triatoma sanguisuga*)
 Mosquitoes
 Scabies mites (*Sarcoptes scabiei*)^a
 Scorpions
 Spiders
 Wasps

ANNULAR ERYTHEMAS

Erythema multiforme
 Annular erythema of infancy
 Acute hemorrhagic edema of infancy
 Erythema annulare centrifugum or gyrate erythema

AUTOIMMUNE OR CONTACT

Animal dander
 Caterpillars and moths (*Lepidoptera*)
 Chemicals
 Cosmetics
 Epoxy resins
 Fish
 Coral
 Jellyfish
 Hedgehog
 Foods
 Medications
 Nickel
 Parabens
 Saliva
 Wood dust

DERMAL HYPERSENSITIVITY REACTIONS

Eosinophilic cellulitis
 Exanthematous drug eruptions

DRUGS

Acetylsalicylic acid^{a,b}
 Allopurinol
 Amoxicillin^a
 Barbiturates
 Cephalosporin antibiotics
 Codeine
 Curare
 Meperidine
 Morphine^a
 Nonsteroidal anti-inflammatory agents (e.g., indomethacin)^{a,b}
 Penicillin^a
 Phenytoin
 Polymyxin B
 Procainamide
 Quinidine
 Iodinated radiocontrast media^b
 Sulfa-derived antibiotics^a
 Sulfonyleureas
 Tetracycline

Thiamine
 Thiazides
 Vancomycin
 Zidovudine

FOOD ADDITIVES, PRESERVATIVES, AND DYES (PSEUDOALLERGIC REACTION)

Azo dyes (e.g., sunset yellow, tartrazine)^a
 Butylhydroxyanisole
 Butylhydroxytoluene
 4-Hydroxybenzoic acid^a
 Sodium benzoate^a
 Sodium metabisulfite

FOOD ALLERGY

Chocolate
 Eggs^a
 Fish^a
 Fresh berries
 Milk^a
 Nuts^a
 Peanuts^a
 Red meat (α -galactosidase)
 Shellfish^a
 Tomatoes

AUTOINFLAMMATORY SYNDROMES

Schnitzler syndrome
 Cryopyrin-associated periodic syndromes (CAPS)
 Familial cold autoinflammatory syndrome
 Neonatal-onset multisystem inflammatory disease (NOMID)
 Muckle-Wells syndrome

INHALANT ALLERGENS

Animal danders
 Mold spores
 Pollens

PHYSICAL AND EMOTIONAL CAUSES

Aquagenic reaction
 Cholinergic stimuli, including emotional stress (psychogenic), exercise, heat
 Cold^a
 Dermatographism^a
 Pressure
 Sunlight
 Sweating
 Vibration

SYSTEMIC DISEASE

Autoimmune thyroid disease
 Celiac disease
 Bullous pemphigoid
 Juvenile idiopathic arthritis
 Kawasaki disease
 Acute rheumatic fever (i.e., erythema marginatum)
 Hypocomplementemic urticarial vasculitis (associated with systemic lupus erythematosus and other connective tissue diseases)
 Serum sickness
 Serum sickness–like reactions
 Common variable immunodeficiency

OTHER

Cutaneous mastocytosis (e.g., urticaria pigmentosa, solitary mastocytoma)

^aMost common.

^bCan cause a pseudoallergic reaction.

nonsteroidal anti-inflammatory drugs [NSAIDs]), and foods; or through inhibition of kinin breakdown by inhibitors of angiotensin-converting enzymes.

Urticaria due to infectious agents is postulated to involve the formation of immune complexes with activation of complement and release of anaphylatoxins or development of IgE antibodies to microbial antigens.^{26,27} There is no reported difference in total serum IgE level, peripheral blood eosinophil count, or IgE-mediated sensitization to aeroallergens in children with acute or chronic urticaria, although aeroallergen sensitization is high in both groups.²⁸

Clinical Manifestations, Differential Diagnosis, and Clinical Approach

Commonly referred to as *hives* or *welts*, urticaria is characterized by the appearance of wheals with or without angioedema. Almost all cases of urticaria in infants and most cases in children are acute and resolve within 6 weeks. Chronic urticaria is diagnosed when urticaria occurs daily or almost daily, with or without angioedema, for >6 weeks.²⁰ Chronic urticaria is further defined as spontaneous or inducible based on whether or not the urticaria can be stimulated by a known source (e.g., cold, pressure, or heat).²⁹

Hallmark clinical features of urticaria include the acute onset of edematous papules and plaques, often with central clearing and surrounded by erythema and associated with pruritus or a burning sensation; they have an evanescent duration of 1–24 hours.²⁹ The size of individual wheals can vary from a few millimeters to several centimeters in diameter. Although classically annular or circular, wheals can assume a bizarre, serpiginous morphology (Fig. 71.1). Occasionally, wheals can have a transient, dusky, violaceous center and can be confused with EM; this variant of acute urticaria has been referred to as *urticaria multiforme* or *acute annular urticaria*.^{30,31} Associated dermatographism, a form of pressure-induced urticaria, often is seen.

Angioedema, defined by edema of the deep dermis and subcutis that classically involves the mucous membranes and acral areas (i.e., hands, feet, and face), occurs in about 5%–10% of children with urticaria. Angioedema usually resolves more slowly than urticaria and can last for up to 72 hours. Patients can have a burning or stinging sensation, but pruritus usually is not prominent.

When urticaria is associated with anaphylaxis, systemic symptoms develop as a result of the release of vasoactive mediators, and swelling of viscera can occur. Systemic symptoms and signs include hoarseness, respiratory distress, emesis, diarrhea, abdominal pain, arthralgia, flushing, syncope, hypotension, and cardiovascular collapse.

The clinical approach to urticaria in children involves establishing a definitive diagnosis of urticaria and identifying a plausible trigger, both of which are usually established by a thorough and pertinent history and physical examination. The history should focus on the frequency and duration of wheals; associated symptoms, such as pruritus or burning; systemic symptoms, such as fever, arthralgia, or respiratory or GI symptoms; and recent medication exposures.²⁹

The physical examination should focus on the morphology, distribution, and progression of wheals, and their transient nature can be documented through serial examinations, with marking of individual wheals if necessary. Angioedema, mucous membrane abnormalities, and specific findings, such as lymphadenopathy, hepatosplenomegaly, and arthritis, should be identified. The differential diagnosis of urticaria includes other annular erythemas such as EM; autoinflammatory syndromes, particularly CAPS; exanthematous drug eruptions; and urticarial eruptions associated with systemic disease, such as systemic lupus erythematosus and juvenile idiopathic arthritis³² (see Box 71.2).

Laboratory Findings and Diagnosis

Skin biopsy can be helpful in equivocal cases or when a diagnosis other than urticaria (e.g., urticarial vasculitis) is suspected. Histologically, urticaria demonstrates dermal edema, dilation of postcapillary venules and lymphatic vessels, and a sparse, mixed perivascular infiltrate with various numbers of eosinophils and neutrophils. An increase in mast cells also may be seen.³³ In some cases, the histology of urticaria can be nonspecific or appear as normal skin.



FIGURE 71.1. Acute urticaria. Migratory, edematous, erythematous plaques with central clearing are associated with edema of the hands and feet.

Routine diagnostic tests are not recommended for cases of acute urticaria.²⁹ Chronic spontaneous urticaria should be evaluated with a differential blood count and either an erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP), and extended testing could be considered based on history.²⁹

Skin prick testing combined with serum-specific IgE testing as clinically indicated can be performed if food or environmental allergy is suspected to be the cause. Oral food challenges and elimination diets, if clinically indicated, should be performed under the guidance of an experienced allergist. Provocation testing may be performed to validate a diagnosis of physical urticaria.

If a specific infectious cause is suspected, directed serologic or microbiologic testing should be performed. Additional laboratory evaluations considered in select cases of urticaria in which an associated systemic disease is suspected may include a complete blood cell count with a differential leukocyte count, ESR, CRP, thyroid function tests, functional autoantibody testing, serum hepatic enzymes, antinuclear antibodies, tryptase, serum immunoglobulins, and complement (e.g., C3, C4, CH50). The basophil histamine release assay is a specialized diagnostic test that can be used to confirm a diagnosis of chronic autoimmune urticaria by detecting IgG antibodies to FcεRI, FcεRII, or IgE or an alternate histamine releasing factor. Imaging studies such as chest or sinus radiography may be considered for select patients.

Management

Management focuses primarily on providing symptomatic relief and eliminating or avoiding the known causes or triggers.³⁴ Factors that can aggravate urticaria include heat, exercise, emotional stress, and medications such as aspirin and NSAIDs.³⁴ Infection should be treated if indicated, although treatment of the associated infection results in improvement of urticaria in only a minority of patients.

The most successful approach to treating the symptoms of urticaria is with medications that block the effect of histamine at the level of its receptor on cutaneous blood vessels. Antihistamines that block H₁ receptors are considered as first-line therapy.³⁴ Hydroxyzine hydrochloride is the most effective of the classic H₁ antihistamines for suppression of the wheal-and-flare response, pruritus, and dermatographism.^{35–37}

Because of significant sedative and anticholinergic effects of the first-generation antihistamines, newer nonsedating H₁ antihistamines such as fexofenadine, loratadine, and cetirizine are recommended as first-line therapeutic agents.³⁴ These agents are as efficacious as first-generation antihistamines and have a longer duration of action. Levocetirizine is safe and effective in reducing the duration of acute urticaria in children as young as 12 months of age.³⁸

Antihistamine therapy should be initiated at the upper end of the recommended dosage range and then gradually increased as needed until symptoms are relieved or until side effects, particularly sedation, become prohibitive. Use of antihistamines at two to four times the recommended dosage by the manufacturer may be required.²⁹ Administering most of the daily dose just before bedtime can minimize daytime sedation.

Alternative or adjunctive therapies should be considered in patients who do not respond to H₁ antihistamines.²⁰ The therapeutic response to the H₁ antihistamines can be improved with the addition of an H₂ receptor antagonist, such as cimetidine or ranitidine, although this effect is limited.²⁰ H₂ receptor antagonists should not be used as monotherapy because of their insufficient H₁ blockade.

Topical corticosteroids are of limited utility in the treatment of urticaria. Use of systemic corticosteroids rarely is indicated in children and is discouraged for prolonged use in cases of chronic urticaria due to concerns about significant adverse effects. A short course of oral corticosteroids usually is reserved for children with severe, unremitting symptoms, especially asthma, laryngeal edema, or anaphylaxis, or when angioedema on the face is particularly severe.²⁰

Rarely, chronic urticaria is severe enough that despite the use of oral antihistamines, additional systemic therapy is warranted. Alternative or adjunctive therapies include mast cell stabilizers, leukotriene antagonists, tricyclic antidepressants (especially doxepin), dapson, cyclosporine, mycophenolate mofetil, and omalizumab.^{20,23,29,34}

Complications and Prognosis

Complications are rare but can result from associated angioedema or anaphylaxis. Most cases of acute urticaria resolve within 1–2 weeks, with or without therapy.³⁹ Approximately one half of cases of chronic urticaria resolve within 5 years, according to one study, although children with inducible urticaria may have more chronic disease.^{40–41}

ERYTHEMA MULTIFORME

Etiologic Agents

Although numerous infections have been reported in association with EM, herpes simplex virus (HSV) is by far the most common, and recurrent EM is significantly associated with HSV infection

(Box 71.3).^{42–45} Unusual presentations include EM associated with neonatal congenital cytomegalovirus infection and hepatitis B or bacille Calmette-Guérin (BCG) immunization and EM associated with

BOX 71.3 Infectious Agents Associated With Erythema Multiforme

BACTERIA

Bartonella henselae
Corynebacterium diphtheriae
Francisella tularensis^a
Legionella pneumophila
Mycobacterium leprae
Mycobacterium tuberculosis^a
Neisseria gonorrhoeae
Proteus mirabilis^a
Pseudomonas aeruginosa
Salmonella spp.^a
Staphylococcus aureus^a
Streptococcus pneumoniae
Streptococcus pyogenes^a
Vibrio parahaemolyticus^a
Yersinia spp.^a

CHLAMYDIA

Chlamydophila psittaci
Chlamydia trachomatis (i.e., lymphogranuloma venereum)

FUNGI

Coccidioides immitis
Histoplasma capsulatum^a

IMMUNIZATIONS

Bacille Calmette-Guérin
Diphtheria and tetanus toxoid
Hepatitis B
Measles
Mumps
Rubella
Poliomyelitis

MYCOPLASMA

Mycoplasma pneumoniae

PARASITES

Trichomonas vaginalis

TREPONEMES

Treponema pallidum

VIRUSES

Adenovirus 7^a
Coxsackieviruses A10, A16, B5
Echovirus 6
Epstein-Barr virus^a
Hepatitis viruses A and B
Herpes simplex viruses 1 and 2^a
Human immunodeficiency virus
Influenza A virus
Measles virus
Mumps virus
Orf virus
Paravaccinia virus^a
Parvovirus B19
SARS-CoV-2
Varicella-zoster virus

^aMost clearly established.

Key Points: Diagnosis and Management of Urticaria

EPIDEMIOLOGY

- Common, occurring in 20%–25% over a lifetime.
- Acute urticaria is typically due to infection, whereas chronic urticaria may be spontaneous or inducible.

CLINICAL FEATURES

- Acute onset of edematous red papules and plaques in a round, annular, or serpiginous pattern.
- Lesions resolve within 24 hours but commonly resolve within 1 hour.

DIAGNOSIS

- Acute urticaria requires no diagnostic testing.
- Workup for chronic urticaria includes a complete blood count (CBC) with differential and erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP) plus additional testing based on history.
- Biopsy may be helpful to exclude other diagnoses.

DIFFERENTIAL DIAGNOSIS

- Urticaria multiforme
- Serum sickness-like reaction
- Viral exanthem
- Exanthematous drug eruption
- Insect bites
- Erythema multiforme

TREATMENT

- Acute urticaria typically does not require treatment; antihistamine can be used for symptomatic relief.
- First-line treatment of chronic urticaria is a second-generation antihistamine (cetirizine, loratadine, fexofenadine) with dose increases based on improvement.
- Refractory chronic urticaria may require additional treatment, including cyclosporine or omalizumab.

Kawasaki disease.^{46–52} Persistent EM, which is rare, has been associated with Epstein-Barr virus (EBV), hepatitis C virus, and influenza virus.⁵³ In children, antibiotics, NSAIDs, and other medications are a common trigger for EM, although in as many as 50% of cases, the etiology is idiopathic.^{54,55}

Epidemiology

Older published reports on EM are confounded by the historical classification of EM as existing on the clinical spectrum with Stevens–Johnson syndrome (SJS)-toxic epidermal necrolysis (TEN); Bastuji-Garin proposed in 1993 that EM was distinct from SJS and TEN.⁵⁶ Some authors continue to use the terms *EM minor* to refer to classic EM with minimal to no mucosal involvement and *EM major* to refer to EM with involvement of one or more mucosa but with less severe involvement than seen with SJS (<10% body surface area [BSA]). In general, only mild oral mucosal involvement is seen in EM, and it is therefore likely that many previously reported cases of EM major represent either reactive infectious mucocutaneous eruption (RIME),⁵⁷ previously called *Mycoplasma pneumoniae*-induced rash and mucositis (MIRM),⁵⁸ or mild SJS.

As such, true epidemiologic data on EM in children are lacking. More commonly a disease of adults 20–40 years of age, an estimated 20% of cases are seen in children and adolescents.^{59–61} There is no clear sex, ethnic, or racial predisposition. Recurrent EM occurs more commonly in the spring.⁴⁵ Human leukocyte antigens B62, B35, B12, and DR53 have been associated with a higher risk for HSV-induced EM, particularly recurrent EM.^{62,63}

Pathogenesis and Pathologic Findings

The pathogenesis of EM is incompletely understood, but evidence implicates a host-specific, cell-mediated immune response (type-IV hypersensitivity) to an antigenic stimulus that targets keratinocytes at the dermal-epidermal junction.^{60,61} Proinflammatory cytokines released by activated mononuclear cells and keratinocytes contribute to keratinocyte necrosis. Although circulating and tissue-bound immune complexes can be demonstrated, they do not appear to play a central role in pathogenesis. HSV antigens and DNA have been documented in skin lesions from patients with EM and are absent in unaffected skin.^{43,64–66}

Histologically, EM demonstrates mild spongiosis, lymphocytic exocytosis, variable papillary dermal edema, a perivascular and interstitial mononuclear infiltrate, and basal vacuolization with liquefactive degeneration of the basal layer of the epidermis; the feature of satellite cell necrosis is a hallmark of EM.⁶⁷ An increased number of eosinophils and/or nuclear dust can also be seen. Subepidermal blisters and necrosis of the entire epidermis can be seen in more severe cases.

Clinical Manifestations, Differential Diagnosis, and Clinical Approach

The onset of EM typically is abrupt. The typical target lesions of EM are characterized by a symmetric, fixed cutaneous eruption of well-demarcated, round, erythematous macules, papules, vesicles, and/or bullae, which most commonly are distributed on the palms, dorsal surfaces of the hands and feet, and extensor surfaces of the arms and legs with relative sparing of the face, trunk, and mucous membranes.⁶⁰

Lesions expand and evolve over several days to assume the classic annular target appearance with a dusky, necrotic center surrounded by a ring of edema and pallor and an erythematous border; arcuate and polycyclic lesions may be seen. Atypical target lesions also can be seen in which either the zone of pallor and edema or the zone of peripheral erythema is lacking or an indistinct border is present (Fig. 71.2). The development of new lesions 72 hours after onset is unusual. The isomorphic phenomenon, which refers to the development of lesions at sites of prior skin trauma, is common and photoaccentuation can be seen. Lesions usually are asymptomatic, although patients may note a burning sensation or mild pruritus.

Prodromal symptoms usually are absent, although low-grade fever, malaise, and upper respiratory tract symptoms can occur. Cervical lymphadenopathy may be seen in association with oral involvement, which occurs in about 25% of cases and usually includes mild, localized erosions involving the gingival, tongue, or buccal mucosa (Fig. 71.3). Rarely, more extensive



FIGURE 71.2. Erythema multiforme. Target lesions manifest as edematous papules with dusky, necrotic centers and erythematous, annular borders.



FIGURE 71.3. Oral aphthae in a child with herpes simplex virus (HSV)-associated recurrent erythema multiforme.

oral mucosal involvement or involvement of other mucous membranes can be seen, but if mucosal involvement is severe, the possibility of SJS or MIRM/RIME should be considered.^{60,61} MIRM/RIME presents with marked mucositis but sparse cutaneous involvement^{57,58} (see Chapter 196).

Lesions of HSV-induced recurrent EM typically develop 10–14 days after the onset of mucocutaneous HSV lesions (usually oral mucosa) or occur after asymptomatic reactivation of HSV.⁶⁸ Recurrent EM has a similar morphology and distribution from episode to episode but can vary considerably in frequency and duration in a given patient.

The initial approach to the child with suspected EM includes discontinuing all nonessential medications and performing a complete examination of the skin and mucous membranes to exclude a serious cutaneous adverse drug eruption, SJS-TEN, or MIRM. Targeted evaluation and management of any possible associated infectious triggers is recommended. The differential diagnosis for EM is broad and includes many other inflammatory skin diseases (Box 71.4).

Laboratory Findings and Diagnosis

Although typically a clinical diagnosis, skin biopsy can help to confirm the diagnosis. There are no typical laboratory abnormalities, although an elevated ESR, moderate leukocytosis, mild elevations of serum hepatic transaminases, or other abnormalities can be seen in some patients and likely reflect on an underlying infection; targeted diagnostic testing should be considered if an infectious etiology is suspected. In cases when

BOX 71.4 Differential Diagnosis of Erythema Multiforme

Acute febrile neutrophilic dermatosis (Sweet syndrome)
 Acute hemorrhagic edema of infancy
 Allergic contact dermatitis (*p*-phenylenediamine in hair dyes, topical triamcinolone)
 Bullous pemphigoid
 Cutaneous small vessel vasculitis
 Exanthematous drug eruption
 Fixed drug eruption
 Henoch-Schönlein purpura
 Linear IgA dermatosis
Mycoplasma pneumonia-induced rash and mucositis (MIRM)/reactive infectious mucocutaneous eruption (RIME)
 Pemphigus vulgaris
 Polymorphous light eruption
 Stevens-Johnson syndrome
 Toxic epidermal necrolysis
 Urticaria
 Urticarial vasculitis
 Viral exanthem, including coxsackievirus A6

HSV infection is suspected, polymerase chain reaction (PCR) or direct fluorescent antigen (DFA) testing can be performed on specimens from primary or recurrent cutaneous lesions.⁶⁷

Management, Empiric Therapy, and Definitive Therapy

Supportive treatment is all that is necessary in most cases. Use of antihistamines (orally), topical corticosteroids, and analgesics (orally and topically for oral mucosal involvement) may be helpful in addressing associated pruritus and pain, respectively. Debridement of necrotic skin is not recommended, and blisters should be left intact when possible. Areas that have eroded, blistered, or crusted should be monitored for infection. Re-epithelialization of denuded areas is facilitated by application of bland emollients, antibiotic ointments, or use of petrolatum-impregnated gauze or hydrocolloid or foam dressings. No controlled, prospective study supports the use of systemic corticosteroids.^{61,69} Oral acyclovir given early for episodes of recurrent HSV-associated EM may lessen the severity of the clinical course.^{68,70} Use of famciclovir for the management of recurrent herpes-associated EM has been reported in a few adults.⁷¹ In adults, the use of immunosuppressive agents such as dapsone, cyclosporine, mycophenolate mofetil and azathioprine, has been reported with variable success in the treatment of recurrent EM.⁶⁷

Complications, Sequelae, and Prevention

Complications are unusual, but secondary bacterial skin infections can occur. If significant oral mucosal involvement is present, scarring may occur. EM is a self-limiting condition that typically resolves within 2–4 weeks. Significant

Key Points: Diagnosis and Management of Erythema Multiforme

EPIDEMIOLOGY

- Associated with many bacterial, fungal and viral infections.
- Herpes simplex virus is the most common inciting infection.
- Medications, especially antibiotics and nonsteroidal anti-inflammatory drugs (NSAIDs), are common triggers.
- The most common affected age group is ages 20–40 years but can occur in infants, children and adolescents.

CLINICAL FEATURES

- Typical target lesions: dusky, necrotic central macule, papule, or bullae surrounded by a ring of edema and pallor and an erythematous border
- Atypical target lesions: dusky, necrotic central macule, papule, or bullae lacking either the zone of pallor and edema or the zone of peripheral erythema; with or without an indistinct border
- Favors acral locations: face, hands, feet
- Variable mild prodrome: low-grade fever, malaise, and upper respiratory tract symptoms
- Mucosal involvement is mild, if present, and usually involves oral mucosa

DIAGNOSIS

- No diagnostic testing is usually required.
- A skin biopsy may be considered if the diagnosis is uncertain.
- Consider targeted evaluation for possible infectious etiologies.

DIFFERENTIAL DIAGNOSIS

- Common EM mimics:
 - Urticaria/urticaria multiforme
 - Stevens-Johnson syndrome (SJS)
 - *Mycoplasma pneumonia*-induced rash and mucositis (MIRM)/reactive infectious mucocutaneous eruption (RIME)

TREATMENT

- Self-limiting
- Supportive care: analgesics, emollients, antihistamines

postinflammatory hyperpigmentation can result, especially in patients with darker skin types, but scarring is unusual unless secondary bacterial infection of the skin has occurred. Recurrent EM usually remits after several years.

Preventive measures for recurrent HSV-associated EM include sun-protective measures such as routine use of sunscreen, use of sun-protective clothing, and avoidance of excess sun exposure. Use of acyclovir or valacyclovir (in adolescents older than 12 years) for 6–12 months can be considered at suppressive doses for children and adolescents with more than three to four recurrences of HSV-associated EM per year (see Chapter 204).

All references are available online at Elsevier eBooks for Practicing Clinicians.

72 Papules, Nodules, and Ulcers

Christine T. Lauren

Papules, nodules, and ulcers are primary lesions of the skin with a variety of infectious and noninfectious causes (Tables 72.1 and 72.2). A *papule* is a raised superficial lesion that is less than 1 cm in diameter whose surface may be smooth, scaly, or hyperkeratotic. A larger (>1 cm), raised, and

often flatter-topped lesion is called a *plaque*. A *nodule* is a solid, palpable lesion that is larger than 1 cm in diameter. An *ulcer* is a loss of skin to the level of the dermis or deeper. An *erosion* is shallower, and the loss of skin is limited to the epidermis.

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